SADIK BHATTARAI

PhD Candidate, Electronics & Information Engineering
Senior Researcher - JUYOUNG BIO, Cor., South Korea
Google Scholar | LinkedIn | YouTube | Instagram

bhattarai.sadik2009@gmail.com
+82-010-6659-8467 | Jeonju, South Korea

RESEARCH PROFILE

Senior Researcher @ JUYOUNG BIO, Cor., specializing in **AI, bioinformatics, molecular dynamics simulations, and AI-driven scientific discovery**, developing **innovative computational solutions** for real-world **biomedical challenges and open science platforms**.

EDUCATION

Integrated Master–PhD in Electronics and Information Engineering — 2021 – 2026
Jeonbuk National University, Jeonju, South Korea

Grade: **[4.43/4.5]**
Dissertation: ***AI-Driven Strategies for Prediction and Design of Anti-Cancer Peptides Targeting HIF-Signaling Pathway***
Advisors: **Prof. Jin-Gyun Chung**

Bachelor of Engineering in Computer Engineering — 2014 – 2018
Everest Engineering and Management College, Pokhara University, Nepal

Grade: [3.5/4]
Thesis: ***Sign Language Translator using OpenCV + MediaPipe***
Advisor: Er. Nabin Thapa

RESEARCH EXPERIENCE

PhD Researcher – Computational Drug Engineering — 2021 – 2026
NSCL Bioinformatics Lab, Jeonbuk National University, South Korea

Senior Researcher — 2026 – Present
https://www.juyoungbio.com/
Jeonju, South Korea

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

Bhattarai, S., Chong, K. T., & Tayara, H. "GAN-ML: Advancing anticancer peptide prediction through innovative Deep Convolution Generative Adversarial Network data augmentation technique." ***Chemometrics and Intelligent Laboratory Systems, 262, 105390.*** **(2025).**

Bhattarai, S., Tayara, H., & Chong, K. T. "Advancing peptide-based cancer therapy with AI: in-depth analysis of state-of-the-art AI models." ***Journal of Chemical Information and Modeling, 64(13), 4941–4957.*** **(2024).**

Bist, P. S., Bhattarai, S., et al. "AntiCPs-CompML: a comprehensive fast track ML method to predict anti-corona peptides." ***BioRxiv (preprint; under review).*** **(2024).**

Bhattarai, S., et al. "ACP-ADA: a boosting method with data augmentation for improved prediction of anticancer peptides." ***International Journal of Molecular Sciences, 23(20), 12194.*** **(2022).**

Conference Papers & Proceedings

Bhattarai, S., et al. "In Silico Computational Method for ACP classification and Peptide Class validation Server in Bioinformatics." ***과학논집 (Dbpia), 48, 31–43.*** **(2023).**

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Manuscripts in Preparation / Submitted / Work on Progress

Bhattarai, S., et al. " De Novo Design of Pharmacokinetically Optimized Cell Penetrating Anticancer Peptides Targeting the CITED2–p300 Regulatory Interface in Hypoxia-Associated Cancer" [Submitted] - (2026).

Bhattarai, S., et al. "Semi-Supervised Generative Design of Antimicrobial Peptides for Combating Multidrug-Resistant Pathogens." [Submitted] - (2026).

Bhattarai, S., et al. "Semi-Supervised Generative Design of Protease Resistant Anti-Acne Peptides for Combating Cutibacterium Acne for Acne Vulgaris" [Work on Progress] - (2026).

Bhattarai, S., et al." Design of Multi-Functional Cell Penetrating Anticancer peptides using ESM-2 Feedback Based Model" [Work on Progress] - (2026).

Bhattarai, S., et al. " Design of Antimicrobial peptides against Gram-Positive, Gram-Negative and Anti-Fungal Strain leveraging Semi-Supervised Probablistic Model and Strain-Similarity Screening."[Work on Progress] - [2026]

AWARDS

Jeonbuk Presidential Research Excellence Award Jeonbuk National University, Jeonju, South Korea	**2026**

PRESENTATIONS & INVITED TALKS

"DCGAN-Based Generative Framework for TNBC-Targeting Peptides with Docking and Stability Validation" 2025 ICROS Annual Conference / ICROS-KROS Honam-Jeju Grand Joint Academic Conference	**2025**
"Advancing Anticancer Peptide Prediction Leveraging DCGAN Network and Motif-Based Authentication" BK21-JIANT WEEK, Jeonbuk National University	**2024**
"ACP-ADA: A Boosting Method with Data Augmentation for Improved Prediction of Anticancer Peptides" (Keynote) Physical Health Conference, Dubai (Remote)	**2023**
"In Silico Computational Method for ACP Classification and Peptide Class Validation" SONSIK Annual Academic Conference, Jeonbuk National University	**2023**

TECHNICAL SKILLS

Machine Learning & Deep Learning

- Supervised, Unsupervised, Reinforcement, and Transfer Learning; Ensemble and Active Learning, Agentic AI
- Neural network architectures: CNNs, RNNs, LSTMs, GANs, Capsule Networks, Transformers
- Hyperparameter optimization: Grid Search, Random Search, Bayesian Optimization
- Model explainability: LIME, SHAP, Dalex; model compression: pruning and quantization

Bioinformatics & Computational Biology

- Generative Model Qualitative and Quantitative Assessment
- Molecular docking and molecular dynamics simulation: AutoDock, HADDOCK, NAMD
- Peptide sequence analysis, motif identification, peptide–protein interaction modeling, Target Identification

AI Frameworks & Libraries

- TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers

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- OpenCV, SpaCy, NLTK, Pandas, NumPy, Matplotlib, Seaborn, Plotly

Programming & Development

- Languages: Python, R, MATLAB, C/C++, Java, SQL, Bash/Shell Scripting, LaTeX
- Tools: Codex/ Claude Code AI and Many More, Jupyter, VS Code, PyCharm, Docker, Git/GitHub, Linux/Ubuntu, Conda/venv, uv package manager, pytest, TensorBoard
- Deployment: Flask, FastAPI

PROFESSIONAL SERVICE

Peer Reviewer

MDPI journals · Elsevier journals · Briefings in Bioinformatics (Oxford Academic) · ACS publications · Nature Communications · PLOS Computational Biology

Academic Collaboration

Advisory Board **Vision & Indochina Intelligence Lab** **THE EEC AI INNOVATION LAB** Building 2, 2nd Floor, Faculty of Science at Sriracha, Kasetsart University, Sriracha Campus	**2026, June- Present**

AFFILIATIONS & COMMUNITY ENGAGEMENT

Foreign Student Leader – JBCIA, Jeonju, South Korea Social Media Content Planning and Support Leader	**2026, June- Present**
Mental Health Ambassador – PRONI Center for Youth Development (Erasmus+) Social media planning and launch, Warsaw, Poland Youth Pass Candidates (Content Ideas and Production) Erasmus Mundas	**2025 – Present**
k-Content Creator – k-Influencer Academy South Korea Promotional Content Creator	**2025 – Present**
Advisor – Fair Action Nepal (FAN) Nagarjun Municipality-09; Kalanki, Kathmandu, Nepal	**2021 – Present**
Ambassador – Red Ginseng Festival, Jinan County Festival Promotion, South Korea	**2023**
Volunteer – Foundazion Un Raggio Di Luce Onlus (FRL / Sarbodaya Nepal) Earthquake Relief Support, Lalitpur, Nepal	**2021**

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LANGUAGES

Nepali	Native
English	Professional working proficiency (Advanced reading & writing)
Korean	Intermediate (Level 3) - KIIP
Hindi	Intermediate

REFERENCES

Prof. Kil To Chong
Professor, Department of Electronics and Information Engineering
Jeonbuk National University, South Korea
kitchong@jbnu.ac.kr

Dr. Hilal Tayara
Assistant Professor, International School of Engineering
Jeonbuk National University, South Korea
hilaltayara@jbnu.ac.kr

Dr. Mapopa Chipofya
Data Scientist, Harvard Medical School
Boston, USA
mchipofya@bwh.harvard.edu